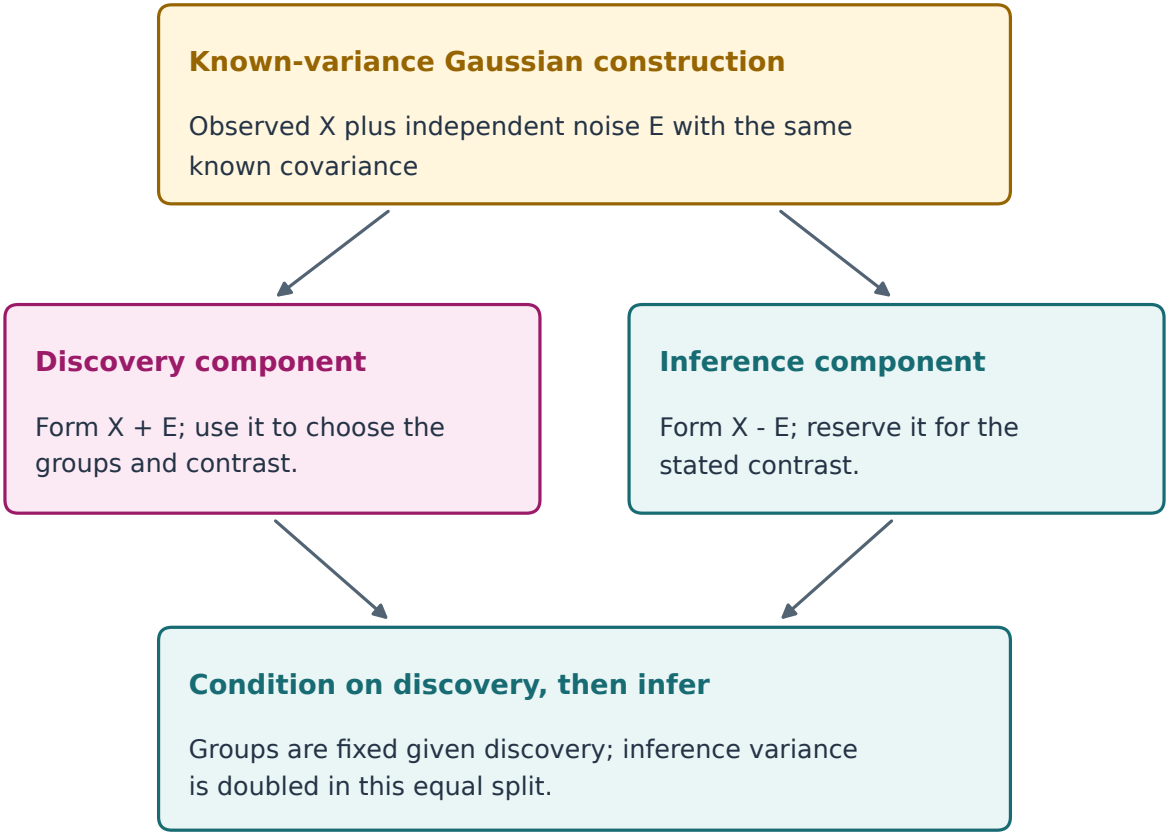
Known-variance Gaussian construction
Observed X plus independent noise E with the same known covariance
Discovery component
Form X + E; use it to choose the groups and contrast.
Inference component
Form X - E; reserve it for the stated contrast.
Condition on discovery, then infer
Groups are fixed given discovery; inference variance is doubled in this equal split.